UNIVERSITY of SOUTH FLORIDA

Muma College of Business

FINAL PROJECT REPORT

AIR POLLUTION AND LIFE EXPECTANCY

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Contents:

1. Introduction 2
2. Research Questions 2
3. Workflow 2
4. Visualization Insights 3
5. Key Findings 9
6. Conclusion 9
7. Additional Research Questions 9

Introduction

Living longer, healthier lives has always been a fundamental human goal. But what truly determines how long we live? While factors like access to healthcare, economic stability, and environmental quality all play a role, one issue has become increasingly clear: air pollution is a silent but powerful force shaping life expectancy worldwide.

This Tableau-based report takes a deep dive into the relationship between life expectancy, air pollution, GDP per capita, and healthcare spending. Using data from trusted sources like the World Bank, World Health Organization (WHO), and Our World in Data, we explore whether our world is on track for longer lives or if environmental damage is holding us back.

Through a series of interactive visualizations, this report tells a story of how air quality and economic factors impact our chances of living a long, healthy life. Our goal is to offer valuable insights for policymakers, researchers, and anyone interested in understanding how the world around us shapes our health and well-being.

Research Questions

1. Are We Living Longer or Shorter Due to Air Pollution? - Life Expectancy
2. Which Countries Have the Longest and Shortest Lifespans?
3. Where in the World Is Air Pollution the Worst?
4. Which Continent Faces the Highest Air Pollution Burden?
5. How does Life Expectancy & GDP Per Capita Evolve Over Time?
6. How does Life Expectancy & Health Expenditures (% of GDP) Evolve Over Time?
7. How does Life Expectancy & Air Pollution Evolve Over Time?
8. Is There Any Link Between Life Expectancy and GDP Per Capita?
9. Is There Any Link Between Life Expectancy and Health Expenditures (% of GDP)?
10. Is There Any Link Between Life Expectancy and Air Pollution?

Workflow

Data Sources:

To investigate the above questions, data was sourced from:

- WHO Global Health Observatory: Life expectancy by country and sex
- World Bank: PM2.5 air pollution (µg/m³), GDP per capita
- Our World in Data: Public health expenditure as % of GDP
- IMF: Global economic indicators (GDP per capita)

Data Preparation and Processing:

The datasets were imported into Tableau, where cleaning and normalization were applied:

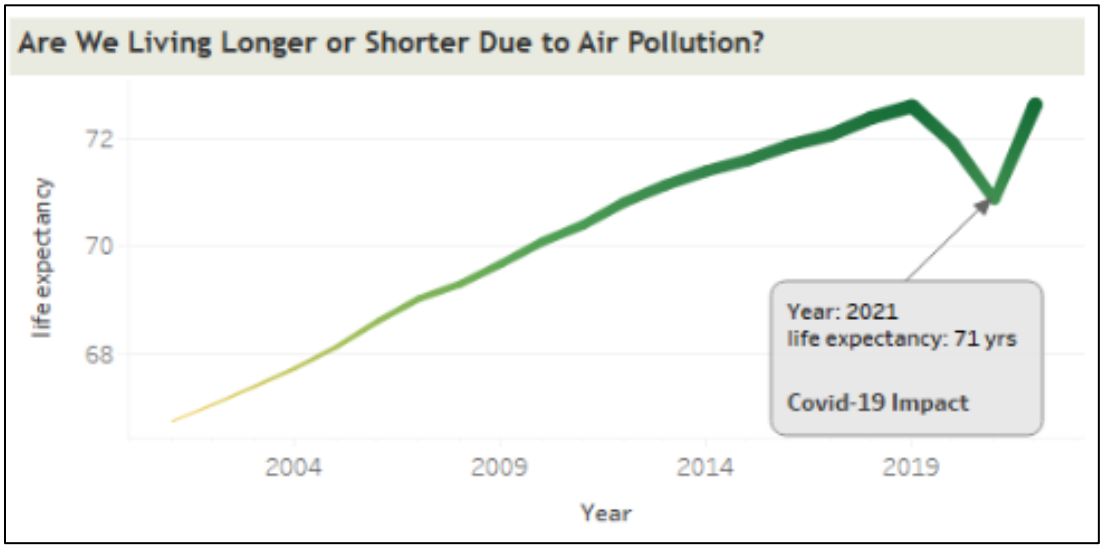
- **Standardization**: Dates were standardized across datasets to ensure consistency in year-over-year comparisons.
- **Joining**: Data was joined using 'Country' and 'Year' as primary keys.
- **Aggregation**: Averages were computed to visualize temporal trends.
- **Outlier Removal**: Extreme values (e.g., life expectancy < 30) were examined for anomalies and retained only if validated.
- **Mapping**: Geospatial data was plotted using latitude and longitude fields for visual impact.

The final dashboard contained linked visualizations to track relationships over time, region, and country.

Visualization Insights

1. Are We Living Longer or Shorter Due to Air Pollution?

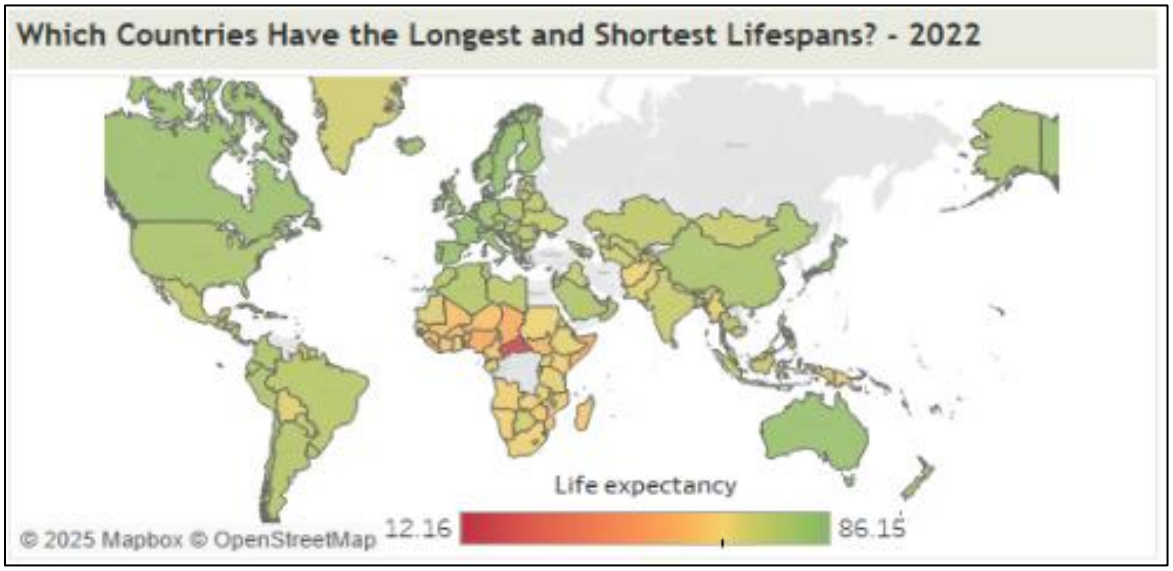
Between 2001 and 2022, life expectancy globally increased from 66.75 to 72.64 years, a gain of nearly 6 years. Meanwhile, average global PM2.5 pollution dropped significantly from 20.64 µg/m³ in 2001 to 10.15 µg/m³ in 2022.



The inverse relationship suggests that efforts to mitigate air pollution might be contributing to increased longevity, though this is influenced by other variables like healthcare quality and economic development.

2. Which Countries Have the Longest and Shortest Lifespans?

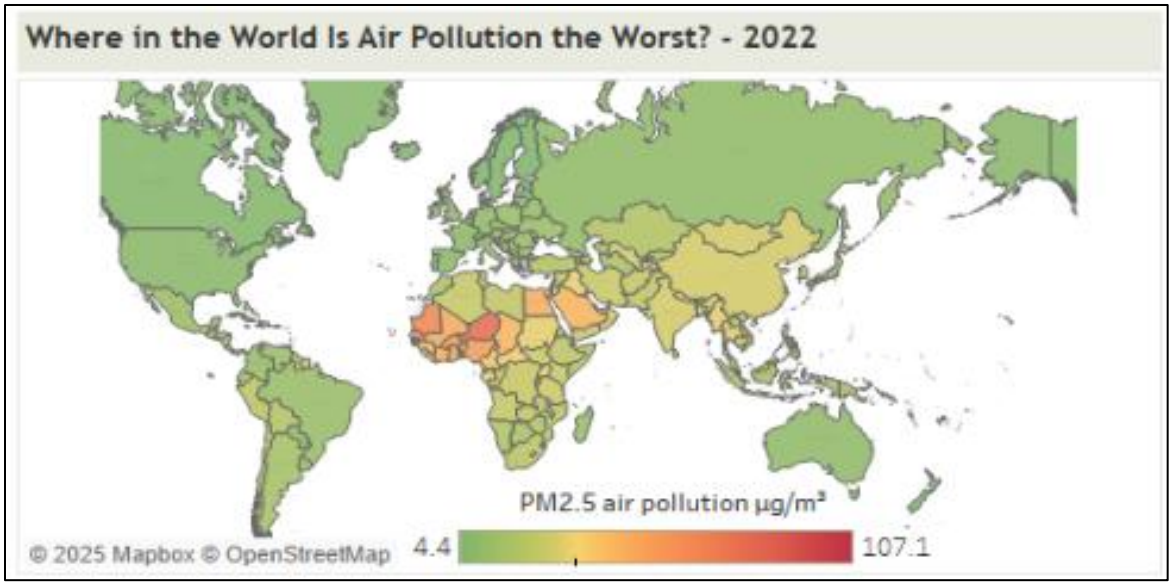
In 2022, Monaco (85.7 years), Japan (84.05 years), and Andorra (84.0 years) recorded the highest life expectancies. Conversely, Central African Republic (18.8 years) and Nigeria (54.1 years) fell at the bottom, highlighting deep health disparities.



Notably, these lower-lifespan countries also had high air pollution levels, poor GDP per capita, and low healthcare investments.

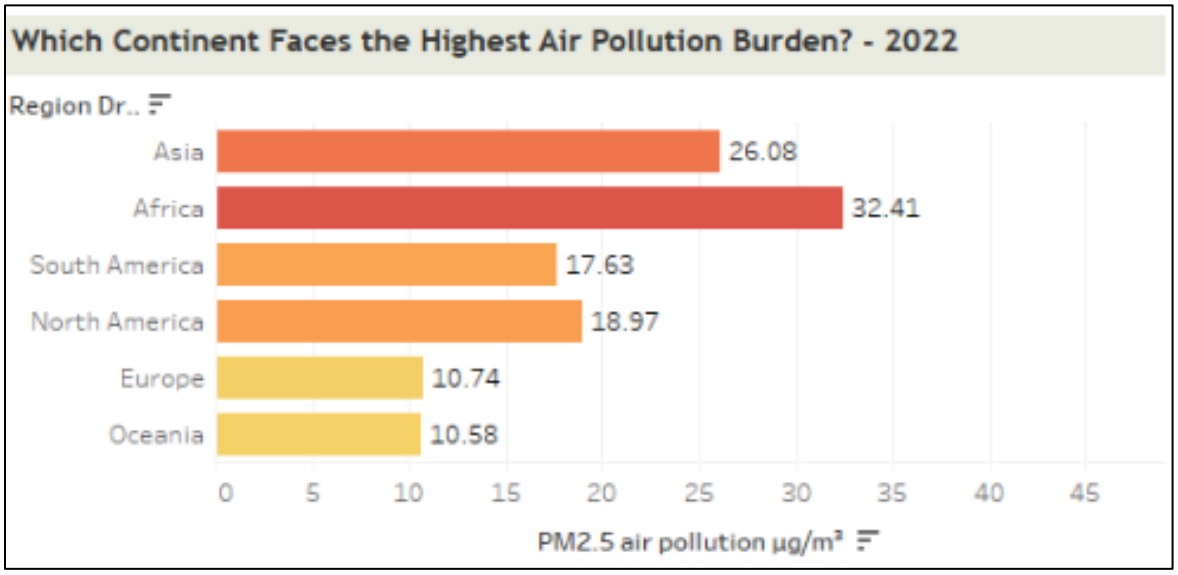
3. Where Is Air Pollution the Worst?

Countries like Mauritania (65.43 µg/m³), Niger (79.85 µg/m³), and Nigeria (54.35 µg/m³) had among the highest PM2.5 levels in 2022. These regions often align with lower life expectancies, indicating a pressing environmental health crisis.



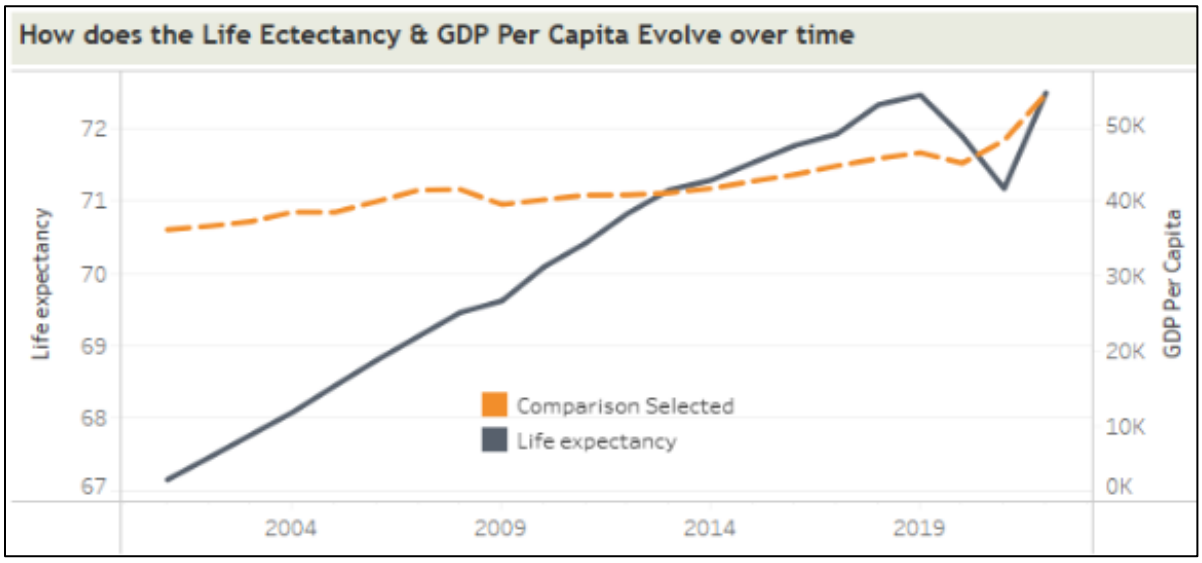
4. Which Continent Faces the Highest Air Pollution Burden?

Africa leads with an average PM2.5 level of 32.4 µg/m³, followed by Asia (26.1 µg/m³). In stark contrast, Europe and Oceania report far lower pollution levels at 10.7 µg/m³ and 10.6 µg/m³, respectively. The regional disparities highlight unequal environmental burdens.



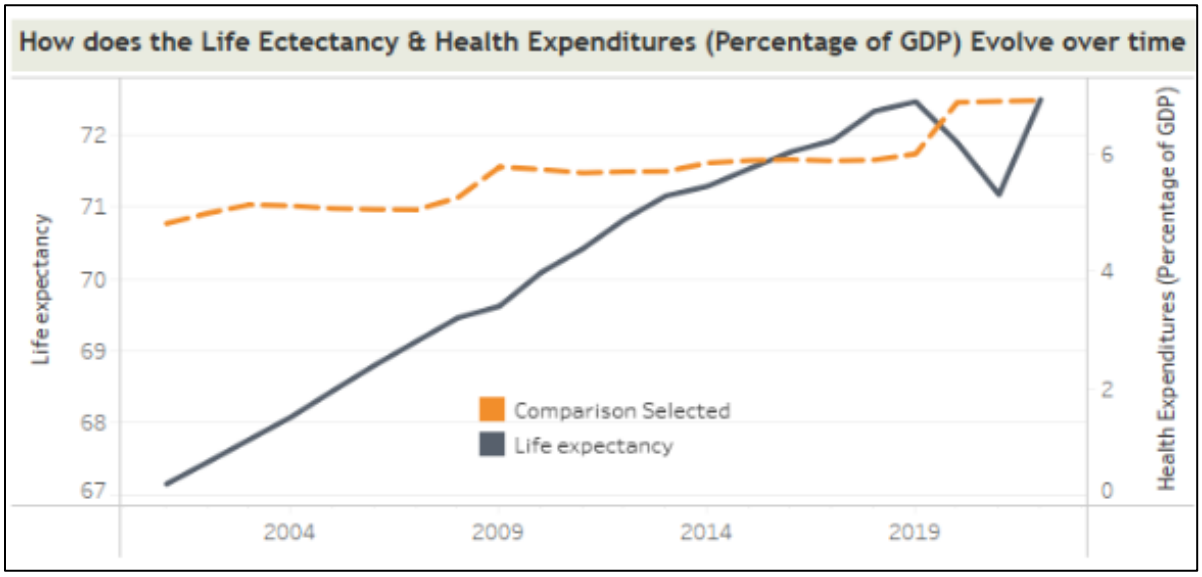
5. Life Expectancy vs. GDP Per Capita Over Time

Between 2001 and 2022, global GDP per capita surged from $36,163 to $54,232, while life expectancy increased from 67.15 to 72.50 years. This consistent rise in both indicators indicates a positive correlation, likely due to better living standards and access to healthcare.



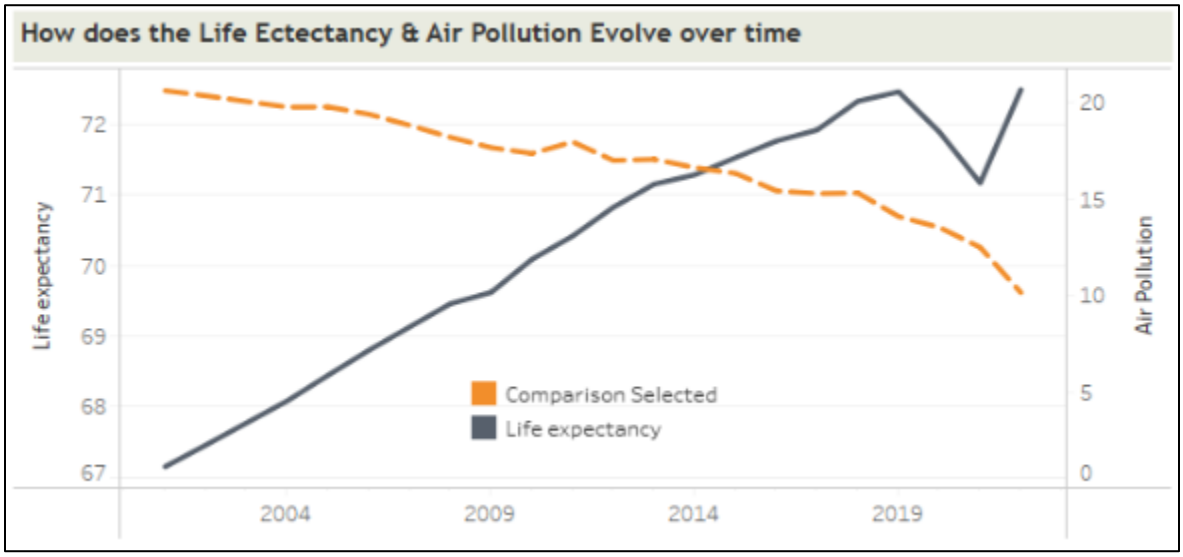
6. Life Expectancy vs. Health Expenditures (% of GDP) Over Time

Health expenditure as a percentage of GDP rose from 4.8% (2001) to 6.9% (2022). In the same period, life expectancy increased by over 5 years. Notably, in 2020, expenditure peaked at 6.87%, possibly reflecting COVID-19 investments, with no corresponding dip in life expectancy, highlighting resilience through investment.



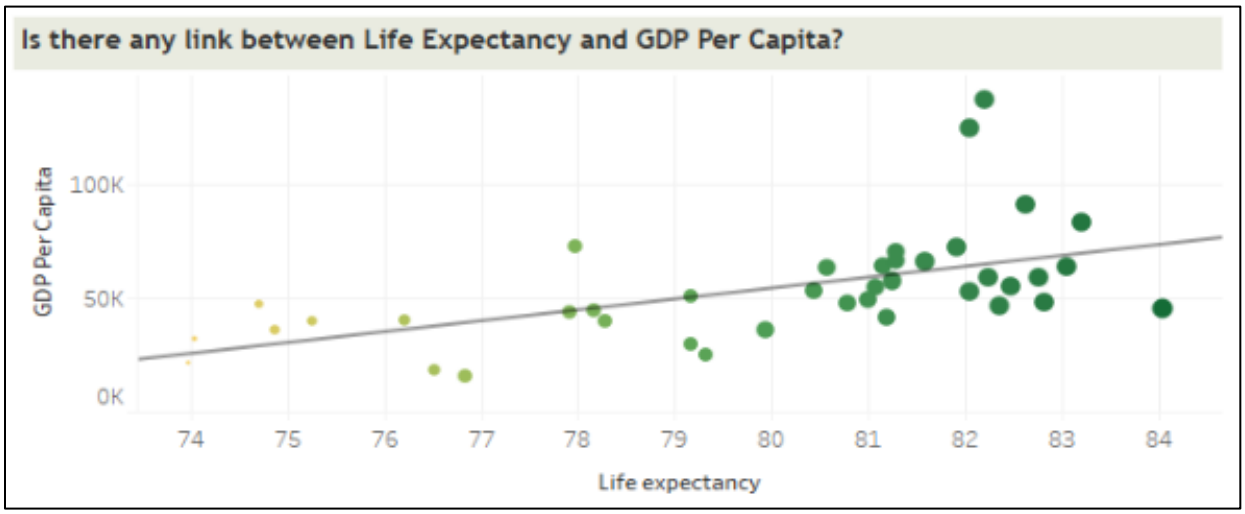
7. Life Expectancy vs. Air Pollution Over Time

PM2.5 pollution fell from 20.6 µg/m³ to 10.15 µg/m³, while life expectancy rose, confirming an inverse relationship. The most significant drop in pollution occurred between 2015 and 2022, suggesting effective environmental policies might be taking effect.



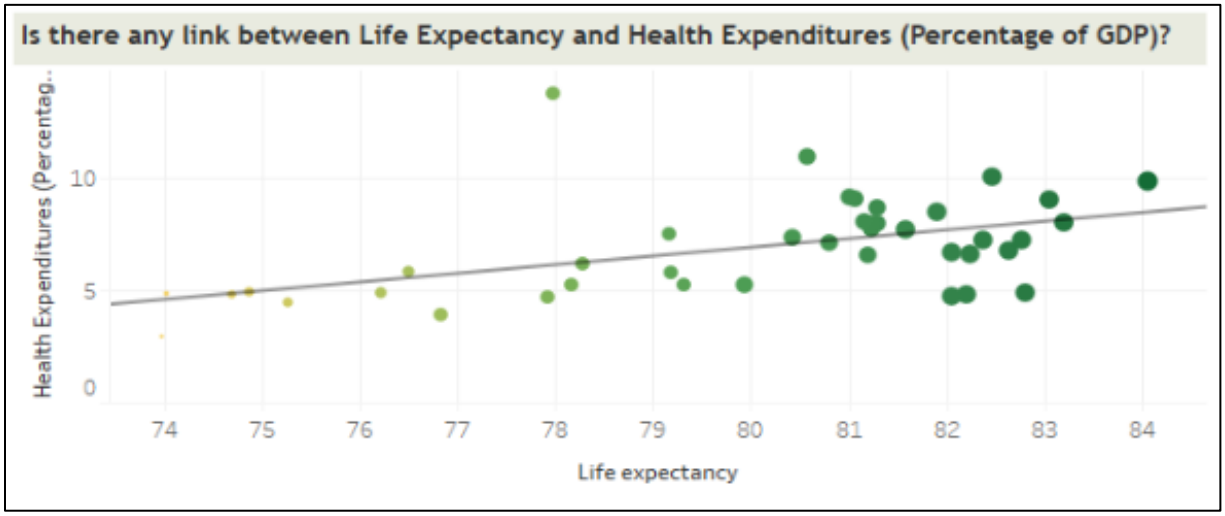
8. Correlation: Life Expectancy and GDP Per Capita

We see a clear positive correlation. For example, Luxembourg (GDP $137k) has a life expectancy of 82.2, whereas Nigeria (GDP ~$2k) sits at 54.1. On average, countries with GDP above $50,000 tend to have life expectancies over 80 years.



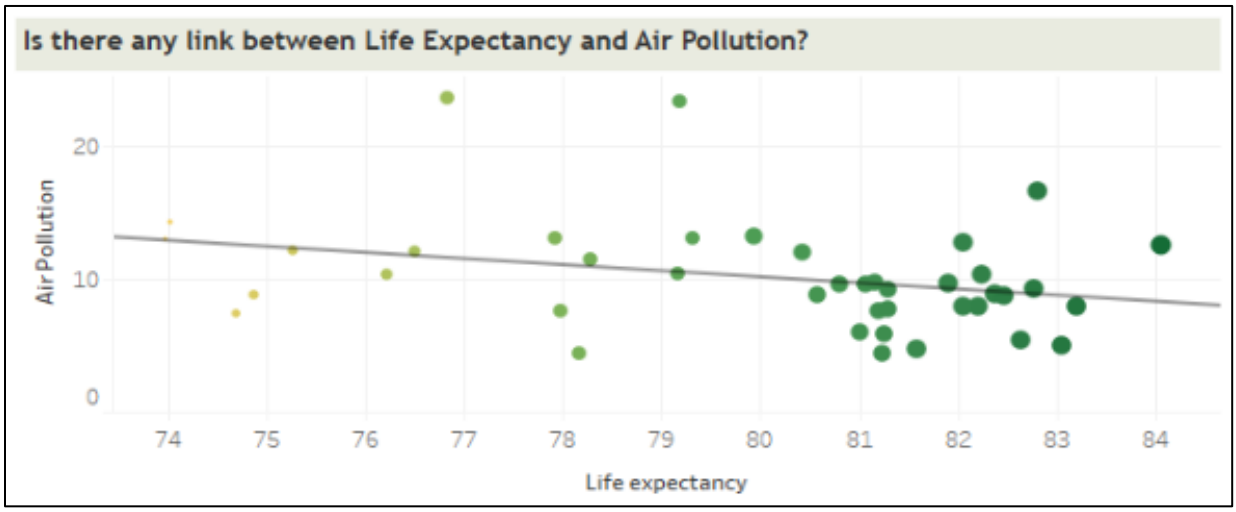
9. Correlation: Life Expectancy and Health Expenditures

The US, spending 13.78% of its GDP on health, has a lower life expectancy (77.97) than Switzerland (8.0%, 83.2 years), showing that high spending doesn't always equate to higher lifespan. However, among 90% of countries, more health investment generally correlates with longer lives.



10. Correlation: Life Expectancy and Air Pollution

Countries with low PM2.5, like Sweden (5.0 µg/m³) and Switzerland (7.9 µg/m³), consistently report higher life expectancies (83+ years), while high-pollution countries like Bangladesh (25 µg/m³) show much lower life spans (~74 years). The data underscores how clean air is a strong predictor of population health.



Key Findings

- Global life expectancy increased steadily over the past two decades.
- Air pollution has declined globally, correlating with improved life expectancy.
- Africa and parts of Asia bear the brunt of air pollution and low life expectancy.
- Countries with high GDP per capita generally have higher life expectancy, though outliers exist.
- Healthcare spending shows a moderate to strong correlation with longevity, especially in mid-GDP countries.
- Air quality is a more direct and consistent predictor of life expectancy than economic indicators alone.
- Environmental interventions appear to be yielding positive health outcomes.

Conclusion

Our analysis tells a simple story: how long we live isn't just about luck, it's shaped by the air we breathe, the healthcare we access, and the strength of our economies. From 2001 to 2022, life expectancy has generally improved, but not for everyone.

In wealthier countries with clean air and strong healthcare, people are living longer. But in poorer regions, where pollution is higher and healthcare is limited, life expectancy still lags. One thing is clear, clean air is a game-changer. The cleaner the air, the longer people live.

For policymakers, the message is clear: investing in clean energy, healthcare, and fair economic growth isn't just good policy, it's a way to help people live longer, healthier lives. Our Tableau dashboard provides an interactive view of these patterns, making it easy to explore how life expectancy connects to air quality, healthcare spending, and economic strength.

Additional Research Questions

1. How does urbanization influence the relationship between pollution and life expectancy?
2. Are there cultural or behavioral factors that mediate the impact of air quality on health?
3. What role does climate change play in altering future projections of life expectancy?

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